Supplementary Material for The Measurement Operator Changes Which Bits Matter: Operator-Conditioned Mixed Precision for Inverse Vision

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1. Supplement Overview

This supplement presents derivations, experimental details, and additional analyses of operator-conditioned precision allocation. The central object is an exact $2^8 = 256$-schedule lattice for an eight-stage ProxSCI-8 solver. The study keeps the learned inverse solver fixed while the calibrated sensing operator changes. All 256 binary precision schedules are evaluated with matched checkpoints, operators, noise realizations, and quantizer settings.

The primary reporting budget is $K = 3$. The tables summarize five-seed behavior, calibration sensitivity, quantizer and noise variations, operator mismatch and transfer, and separately measured native INT8/FP16 timing. Transfer summaries are reported as ranges, not as confidence intervals. External published comparison values are distinguished from measurements in this study.

2. Notation and Evaluation Objectives

We study

$$y = Hx^\star + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \Sigma), \tag{1}$$

and define the whitened normal operator

$$M_H = H^\top \Sigma^{-1} H. \tag{2}$$

For high-precision execution $\hat{x}_{\mathrm{high}}$ and mixed-precision schedule z producing $\hat{x}_z$, the primary numerical discrepancy is

$$D_H(z) = \frac{1}{m} \left\| \Sigma^{-1/2} H(\hat{x}_{\mathrm{high}} - \hat{x}_z) \right\|_2^2. \tag{3}$$

D_H is teacher-relative and operator-visible. It is not ground-truth reconstruction error and is blind to $\ker(H)$. Reconstruction quality is tabulated separately. We do not report a quantitative observed/null-space decomposition or per-schedule D_H–PSNR correlation.

For the all-low execution $\hat{x}_{\mathrm{low}}$ and matched stage-j promotion $\hat{x}_{j\uparrow}$,

$$q_j = \hat{x}_{\mathrm{low}} - \hat{x}_{j\uparrow}. \tag{4}$$

The calibration-distribution susceptibility is the uncentered second moment

$$K_j^{\mathrm{cal}} = \mathbb{E}[q_j q_j^\top] = \mathrm{Cov}(q_j) + \mu_j \mu_j^\top. \tag{5}$$

With $K_j^{\mathrm{cal}} \approx U_j \Lambda_j U_j^\top$, OSM scores stage j for operator H by

$$s_j(H) = \mathrm{Tr}(M_H K_j^{\mathrm{cal}}) = \left\| \Sigma^{-1/2} H U_j \Lambda_j^{1/2} \right\|_F^2. \tag{6}$$

The score is a calibrated, operator-conditioned susceptibility proxy, not a demonstrated exact marginal promotion gain. The signed alignment term in the identity below can change the gain ordering.

Table 1. Summary of principal measured results by mechanism, allocation quality, reconstruction quality, robustness, and deployment regime.

Evidence	Result	Regime
Operator reordering	WORI 0.29; full $\tau = 0.80$	Five seeds
Frozen geometry	$\tau = 0.86$; boundary MAE 3.08°	Isospectral/controlled path
Static regret	$6.34\% \rightarrow 2.70\%$	$K = 3$
Exact agreement	$\approx 58\%$	$K = 3$
Calibration size	$\tau = 0.64$–0.84	$N = 256$–$2{,}048$
QAT/PTQ	2.3–2.7% OSM regret	Quantizer evaluation
Mismatch	2.5–15%	Calibrated to combined
Generalization	$\tau = 0.64$–0.84	Held-out/extrapolation/deblur
SCI quality	35.29 dB OSM	P1
Quantizer robustness	35.26 dB OSM (max-abs PTQ)	P1 sweep
Native timing	16.4 ms p50, 1.44×	P4

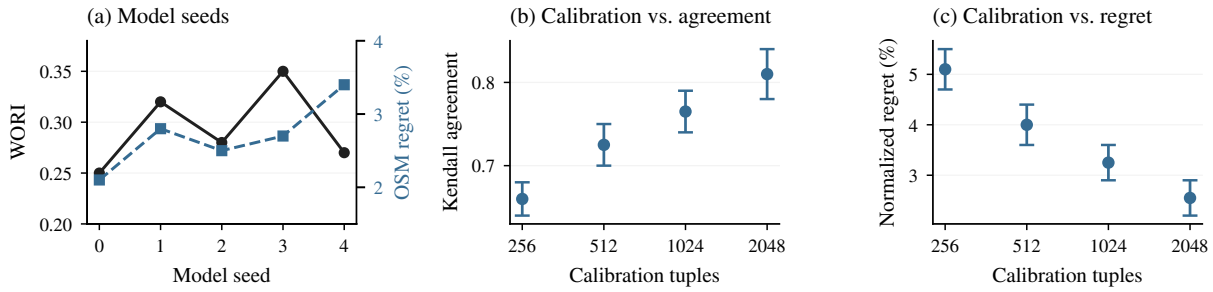


Figure 1. **Five-seed variability and calibration sensitivity.** (a) Per-seed WORI and OSM regret. (b,c) Range endpoints from the calibration-size study (Table 5), shown as whiskers rather than trajectories or confidence intervals.

3. Exact Singleton Gain and Residual Alignment

Let $e = \hat{x}_{\text{high}} - \hat{x}_{\text{low}}$ and $r_j = \hat{x}_{\text{high}} - \hat{x}_{j\uparrow} = e + q_j$, the discrepancy remaining after promoting stage j. Direct expansion gives

$$\Delta_j(H) = D_H(\text{low}) - D_H(j\uparrow) = \frac{q_j^\top M_H q_j - 2 r_j^\top M_H q_j}{m}. \tag{7}$$

Define the symmetric calibration cross moment

$$C_j^{\text{cal}} = \mathbb{E}_{\text{cal}}\left[\frac{q_j r_j^\top + r_j q_j^\top}{2}\right]. \tag{8}$$

The expected singleton gain is therefore

$$g_j(H) = \frac{\operatorname{Tr}(M_H K_j^{\text{cal}}) - 2\operatorname{Tr}(M_H C_j^{\text{cal}})}{m} = \frac{s_j(H) - 2\operatorname{Tr}(M_H C_j^{\text{cal}})}{m}. \tag{9}$$

OSM ranks stages using the nonnegative self-energy term $s_j(H)$ and omits residual alignment. Under an additive perturbation model $\hat{x}_{\text{low}} = \hat{x}_{\text{high}} + \sum_k d_k$ with $q_j = d_j$, the omitted contribution depends on M_H-weighted cross-stage products

Table 2. Five-seed mechanism and regret measurements. Each row is one model seed; regret is normalized operator-visible regret at $K = 3$.

Seed	WORI	Frozen τ	Full τ	Global regret	OSM regret	Boundary MAE
0	0.25	0.84	0.79	5.9%	2.1%	2.2°
1	0.32	0.90	0.85	6.8%	2.8%	3.4°
2	0.28	0.86	0.80	6.1%	2.5%	2.6°
3	0.35	0.89	0.84	7.2%	2.7%	3.1°
4	0.27	0.81	0.74	5.7%	3.4%	4.1°
Mean	0.29	0.86	0.80	6.34%	2.70%	3.08°

$\mathbb{E}[d_k^\top M_H d_j]$ for $k \neq j$. If these vanish, the expected gain reduces to $s_j(H)/m$, sufficient for the same ranking. This condition is not assumed to hold exactly for the nonlinear solver; we do not evaluate a score that explicitly includes the cross moment.

4. Static-Policy Regret Consequence

Let $\mathcal{Z}_C$ be the feasible schedule set at cost C and $z_H^\star = \arg\min_{z \in \mathcal{Z}_C} D_H(z)$. If two operators have distinct unique optima with positive non-optimality gaps γ_i, any deterministic static schedule incurs

$$\max_i \left[D_{H_i}(z) - D_{H_i}(z_{H_i}^\star) \right] \geq \min(\gamma_1, \gamma_2) > 0. \tag{10}$$

This is a finite-set consequence of distinct optima, not a quantitative guarantee when schedule gaps are near zero. The $K = 3$ regret measures the empirical magnitude of the static-policy cost.

5. Geometry, Trajectory, and Invariance Controls

Changing H can affect both physical geometry and the learned reconstruction trajectory. Writing $K_j(H) = K_j^{\text{cal}} + \Delta K_j(H)$ gives

$$s_j^*(H) = \text{Tr}(M_H K_j(H)) = \text{Tr}(M_H K_j^{\text{cal}}) + \text{Tr}(M_H \Delta K_j(H)). \tag{11}$$

A frozen-perturbation experiment holds q_j fixed while changing M_H; a full-system experiment regenerates measurements and reruns the solver.

5.1. Five-seed mechanism result

Across the five independent model seeds, operator-induced reordering varies from 0.25 to 0.35, and full-system ranking agreement varies from 0.74 to 0.85. The corresponding global-static regret exceeds OSM regret in each seed, with differences ranging from 2.3 to 4.5 percentage points. Table 2 lists the per-seed values and their means.

5.2. Continuous crossings

Across the controlled paths, frozen-score Kendall correlation is approximately 0.86 with seed SD 0.03–0.04; Spearman correlation is 0.89–0.92, top-3 overlap is 0.82–0.86, pairwise flip precision is 0.87–0.91, and pairwise flip recall is 0.82–0.87. Paths contain typically 3–6 meaningful transitions over $0°$–$90°$, but not every path has the same number. Boundary MAE is $3.1°$, with occasional 6–8° misses.

5.3. Same-normal invariance

If $M_{H_1} = M_{H_2}$ and the same K_j^{cal} is used, $s_j(H_1) = s_j(H_2)$ exactly. The measured FP32 unit test has maximum relative score error $\leq 10^{-6}$ and 100% identical OSM schedules under deterministic tie-breaking.

5.4. Score-factor controls

The matched controls separate the uncentered second moment into covariance and mean-energy terms and compare full M_H against diagonal, isotropic, and unwhitened normal-operator surrogates. Other diagnostic controls include identity/isotropic sensing, stage-index selection, content-only routing, and randomized susceptibility bases; individual quantitative comparisons for these controls are not reported here.

Table 3. Operator controls and the confound each control addresses.

Control	Purpose
Same latent/model/quantizer	Remove scene/model confounding
Matched spectrum/rank	Control spectral conditioning
Matched whitened SNR	Control measurement-noise scale
Matched observable energy	Control measured-subspace energy
Matched FP quality/residual	Control realized inverse difficulty
Same normal operator	Representation invariance
Frozen q_j	Isolate direct geometry
$H = I$/isotropic	Test necessity of anisotropy
Random basis	Destroy susceptibility alignment

Table 4. Arithmetic regimes. P1 and P4 are deliberately kept separate in all interpretation.

Regime	Low/high	Role
P1	W8A4/W8A8	Primary activation policy
P2	W4A4/W8A8	Whole-stage evaluation
P3	W4A8/W8A8	Weight-only evaluation
P4	INT8/FP16	Native deployment

6. SCI Forward Model and Operator Families

For eight-frame SCI,

$$y_p = \sum_{t=1}^{8} c_{p,t} x_{p,t} + \varepsilon_p, \tag{12}$$

with $c_{p,t} \geq 0$. The implemented adjoint is numerically checked using random test vectors. Operator families comprise algebraic isospectral paths, positive temporal-code paths, ordinary SCI masks, and mismatch perturbations. The main physical experiments use matched scene/model/noise tuples.

7. Data Splits, Architecture, and Training

Training uses DAVIS2017 clips [Pont-Tuset et al. 2017]. The study contains 2,048 disjoint 256×256 calibration clip-measurement pairs, 1,024 held-out validation pairs, and 64 fixed latent clips for the controlled test. Standard simulated SCI sequences are Kobe, Traffic, Runner, Drop, Crash, and Aerial; public real observations include Domino and Water Balloon from the established SCI benchmark [Qiao et al. 2020]. Training seeds are 0–4.

ProxSCI-8 has eight unfolded stages with a data-consistency update and shared-weight learned proximal block. The proximal network uses a 3×3 16→64 stem, eight residual blocks with 64-channel 3×3 convolutions, exact GELU, residual scale 0.1, and a 64→8 head, without BatchNorm. Training uses AdamW, learning rate 2×10^{-4} cosine-decayed to 2×10^{-6}, weight decay 10^{-5}, and gradient clipping 1.0. The high-precision checkpoint is selected independently of the later precision-policy evaluation.

8. Quantization and Stage Interventions

Quantization is symmetric with zero point 0 and round-to-nearest-even. Weights use per-output-channel max-absolute scales; activations use frozen per-stage/per-layer calibration scales from the 2,048 calibration clips. The primary causal study does not use test-time dynamic absmax. Simulated QDQ accumulates in FP32 unless accumulator precision is itself the variable. Activation precision shows stronger operator-conditioned reordering (WORI about 0.25–0.34) than weight precision (0.14–0.20).

Table 5. Calibration-size sensitivity on held-out operators.

Calibration N	Kendall τ	Mean regret
256	0.64–0.68	4.7–5.5%
512	0.70–0.75	3.6–4.4%
1,024	0.74–0.79	2.9–3.6%
2,048	0.78–0.84	2.2–2.9%

9. Susceptibility Rank and Calibration Stability

Candidate ranks are $\{8, 16, 32, 64, 96\}$. Most selected ranks are 32 or 64, with occasional 16 or 96. Rank 8 underfits the reported sweep, while rank 96 adds little additional value. The selection rule retains at least 97% of second-moment energy and at least 95% of comparable pairwise ordering.

Re-estimating susceptibility from separated reference sub-pools gives cross-reference stage-score Kendall τ of approximately 0.75–0.88; most crossing shifts are 1–4°, with occasional 5–8° shifts, and held-out regret changes by about 0.3–1.2 percentage points. These measurements quantify dependence on the calibration-operator distribution.

10. Exact 256-Schedule Lattice and Regret

Every scene/operator/noise/model tuple contains 256 schedule IDs. Schedule rows share one checkpoint, operator/noise realization, and frozen scales. Operator-visible and reconstruction-quality optima are computed separately. The exact lattice is evaluated for every equal-cost $K = 0, \ldots, 8$ and measured-latency feasible set.

For schedule z and oracle $z_H^\star$,

$$R_{\mathrm{raw}} = D_H(z) - D_H(z_H^\star), \qquad R_{\mathrm{norm}} = 100 \frac{D_H(z) - D_H(z_H^\star)}{D_H(z_{\mathrm{all-low}}) - D_H(z_H^\star)}. \tag{13}$$

Normalized regret is the reported comparison scale at $K = 3$; it is calculated from raw D_H, which is not included in the aggregate ledger. A fixed equivalence tolerance avoids dividing by an effectively zero low-to-oracle gap. The same fixed equivalence rule is used across all policies on a tuple, ensuring a common evaluated population.

Table 6. Partition of the exact binary lattice by equal-cost budget. The counts sum to all 256 schedules; the D_H oracle is recomputed independently within each budget.

K	0	1	2	3	4	5	6	7	8
Schedules	1	8	28	56	70	56	28	8	1

Table 8. Explained variance of exact-lattice discrepancy.

Seed	Additive R^2	+ pairwise R^2
0	0.82	0.92
1	0.87	0.94
2	0.84	0.93
3	0.79	0.90
4	0.86	0.95
Mean	0.84	0.93

Table 7. Measured exact-lattice comparison at the primary equal-cost budget $K = 3$. The baseline ordering is heterogeneous across regret and overlap metrics.

Method	Mean regret (%)	p90 (%)	Top-K Jaccard
Random matched-cost	18.7	35.1	–
Early-high	13.4	24.5	0.48
Late-high	12.1	22.8	0.50
Quant-MSE	9.2	17.1	0.58
Operator-difficulty scalar	8.2	16.6	0.56
Global static	6.3	13.2	0.62
Gradient	6.0	12.7	0.65
H-aware gradient	5.0	9.9	0.70
Hessian/Fisher	4.8	9.4	0.67
InfoQ-style	5.1	9.0	0.71
Intervention	4.4	8.8	0.72
Exact Shapley	3.8	7.4	0.76
Greedy re-evaluation	3.3	6.9	0.78
OSM	2.7	6.4	0.75
Per-operator calibration	1.7	3.5	0.84
Exact D_H oracle	0.0	0.0	1.00

The experiment enumerates 256 binary schedules; Table 7 presents the $K = 3$ operating point. For reference, uniformly random three-stage selection has expected top-3 Jaccard 0.259 against a fixed oracle schedule. The random-policy measurement in Table 7 uses its original reported regret; a Jaccard value is not assigned without its sampling distribution.

OSM exactly matches the oracle schedule in approximately 58% of the primary $K = 3$ test cases. The held-out summary is $\tau \approx 0.80$ with seed SD about 0.05, top-2 Jaccard 0.76–0.79, top-3 Jaccard 0.73–0.76, mean normalized regret 2.6–2.8%, median regret 1.8–2.1%, p90 6.2–6.6%, worst held-out operator 11–12%, and 83–88% of operators within 5% of the oracle.

11. Interaction Analysis

The pairwise model materially improves explained variance, but the residual remains nonzero. This is why OSM is interpreted as a transferable stage-ordering proxy rather than an exact additive decomposition.

12. Baseline Implementations

The score-based comparators use the declared calibration tuples, precision pair, and cost table, but they differ in test-operator access. The distinction matters because OSM computes a score for a new calibrated H without measuring candidate schedule outcomes on that operator. Table 9 records the access model used in interpreting the reported $K = 3$ comparison; a matched-cost test-H search frontier is not part of the study. The test lattice evaluates policies and does not tune their parameters.

Table 9. Test-operator access for the allocation comparison. "Calibration only" means that candidate schedules are selected from the fixed calibration distribution rather than by test-H schedule-outcome evaluation.

Policy	Test-operator information during selection
Global static	None; fixed schedule
Gradient; Hessian/Fisher; InfoQ	Calibration only
Intervention; greedy; Shapley	Calibration schedule outcomes
H-aware gradient	Test H in gradient objective
OSM	Test H analytically in $s_j(H)$
Per-operator calibration	Test-H schedule outcomes
Exact oracle	Test-H full schedule outcomes (evaluation)

Gradient and H-aware gradient. The generic gradient score uses mean absolute first-order reconstruction-loss sensitivity; the H-aware variant replaces the objective with D_H.

HAWQ-style Hessian/Fisher. The curvature baseline uses trace-weighted sensitivity with fixed Rademacher probes on a stratified calibration subset [Dong et al. 2020].

InfoQ-style. The information baseline uses observer stages $\{2, 4, 6, 8\}$, fixed PCA/projection directions, and normalized downstream information change [Akbulut et al. 2026].

Intervention, greedy, Shapley. Singleton intervention promotes one stage; greedy recomputes marginals; exact Shapley uses all 256 schedule outcomes on calibration.

Controls not compared quantitatively. Operator-cluster routing, scene-only routing, adaptive depth, and compact-model interventions were not evaluated as matched quantitative baselines. The reported comparisons therefore concern the tabulated methods.

13. Evaluator and Statistical Protocol

PSNR uses clipped $[0, 1]$ predictions and float64 per-frame MSE; SSIM uses an 11×11 Gaussian window with $\sigma = 1.5$, $K_1 = 0.01$, $K_2 = 0.03$, population covariance, and reflect padding. LPIPS, when reported, uses AlexNet with grayscale repeated to RGB. Temporal-gradient error is the primary temporal metric. Comparisons are paired on identical scene/operator/noise tuples. Kendall uses τ_b.

The comparisons in this supplement are descriptive (means, p90 values and ranges); no confidence intervals or hypothesis tests are reported. Small rounded differences are not interpreted as statistically significant.

14. Scene/Operator Variance and Noise Robustness

Operator effects are material, but scene and interaction effects remain substantial.

15. Quantizer, QAT, and Precision-Object Robustness

QAT raises OSM PSNR to 35.35 dB and lowers normalized regret to 2.3%, with ranking agreement $\tau = 0.77$ (Table 13).

Table 10. Descriptive endpoint registry. Differences are arithmetic differences of rounded means and are not inferential tests.

Endpoint	Estimate	Comparison	Interpretation
WORI	0.29	–	Material operator-induced reordering
Frozen/full τ	0.86 / 0.80	-0.06	Trajectory effects are material
OSM/global regret	2.7 / 6.3%	-3.6 pp	Lower oracle-relative regret
OSM/H-gradient	2.7 / 5.0%	-2.3 pp	Geometry overlap adds value
OSM/greedy	2.7 / 3.3%	-0.6 pp	Higher overlap, higher reported regret
OSM/per-operator	2.7 / 1.7%	$+1.0$ pp	Fixed-sensor search remains an upper baseline
Boundary MAE	3.08°	–	Nonzero transition error
Native speedup	$1.44\times$	vs FP16	Mixed precision at a preselected timing budget

Table 11. Observed variance decomposition.

Component	Observed range
Operator	28–38%
Scene	21–31%
Scene$\times$operator	18–29%
Model seed	2–8%
Residual	10–18%

Table 12. Noise robustness. WORI is non-monotonic across the evaluated SNR levels.

SNR	WORI	OSM regret
20 dB	0.23	3.7%
25 dB	0.28	3.0%
30 dB	0.30	2.5%
35 dB	0.27	2.7%
40 dB	0.31	2.3%

Table 13. PTQ/QAT robustness. PSNR values are compared within this quantizer sweep.

Setting	Uniform PSNR	OSM PSNR	WORI	τ	Regret
PTQ max-abs	34.76	35.26	0.29	0.80	2.7%
PTQ percentile	34.82	35.30	0.26	0.78	2.6%
PTQ MSE clip	34.85	35.28	0.28	0.82	2.5%
QAT	34.99	35.35	0.23	0.77	2.3%

Table 14. Activation and weight precision.

Object	WORI	OSM τ
P3 weights W4A8/W8A8	0.14–0.20	0.68–0.75
P1 activations W8A4/W8A8	0.25–0.34	0.77–0.85
P2 whole-stage W4A4/W8A8	0.21–0.30	0.73–0.82

16. SCI Reconstruction Quality

The SCI reconstruction comparison in Table 16 reports 35.29 dB for OSM (34.74 dB uniform-low), whereas the max-abs PTQ robustness summary in Sec. 15 reports 35.26 dB (34.76 dB uniform-low). The quantizer sweep and the quality comparison are

Table 15. Calibration distribution and susceptibility-reference stability.

Quantity	Observed range
Cross-reference stage-score τ	0.75–0.88
Typical crossing shift	1–4°
Occasional crossing shift	5–8°
Held-out regret change	0.3–1.2 pp

Table 16. Standard SCI reconstruction quality. Published EfficientSCI-S and Q-SCI anchors [Wang et al. 2023, Cao et al. 2024] are marked with *; the remaining rows use the evaluator described in Sec. 13.

Method	Precision	PSNR (dB)	SSIM
EfficientSCI-S*	FP	35.51	0.970
Q-SCI 4-bit*	4-bit	34.69	0.963
Q-SCI 8-bit*	8-bit	35.57	0.969
Uniform low	W8A4	34.74	0.964
Global static mixed	W8 A4/A8	35.06	0.966
H-aware gradient	W8 A4/A8	35.16	0.967
OSM	W8 A4/A8	35.29	0.968
Per-operator static	W8 A4/A8	35.23	0.968–0.969
Uniform high quantized	W8A8	35.43	0.969

Table 17. Generalization ranges across seeds.

Regime	Kendall τ	OSM regret
Held-out mask seeds	0.78–0.84	2.2–2.9%
Held-out operator intervals	0.73–0.80	2.6–3.6%
Extrapolative operators	0.64–0.72	3.6–5.1%
Non-blind deblurring	0.70–0.77	3.7–4.8%

evaluated and summarized separately; comparisons are made within each table. Per-operator calibration has lower D_H regret but slightly lower PSNR; the metrics assess different reconstruction properties.

17. Held-Out Transfer and Second Inverse Task

Regret is smallest on held-out masks and increases under operator extrapolation. Deblurring uses a separate forward operator and does not share SCI stage identities. A smaller EfficientSCI-S transfer study gives WORI about 0.18–0.25, OSM regret about 3–4%, and global-static regret about 6–8%, as summarized in the submitted supplementary analysis. ProxSCI-8 remains the primary exact-lattice evidence.

18. Operator Mismatch and Calibration

The physical/oracle operator, nominal operator, and calibrated operator are distinguished in the mismatch experiments. Under combined forward-operator mismatch, OSM regret rises to 11–15% (Table 18). Recalibration recovery and an uncertainty-triggered fallback are not evaluated here.

19. Observed and Null-Space Diagnostics

D_H is blind to $\ker(H)$, so an observable/null-space decomposition would be needed to measure damage not seen through the forward operator; we do not report such a decomposition. Equal image-space perturbation norms can have very different D_H depending on alignment with observable directions; conversely, a visually harmful null-space perturbation can leave D_H nearly unchanged. This is why high-precision teacher preservation and ground-truth reconstruction metrics remain separate

Table 18. Operator mismatch. Schedule distance follows the original study's stage-change convention (distinct from binary Hamming distance).

Condition	Schedule distance	OSM regret
Calibrated	0.3–0.6	2.5–3.2%
Small gain error	0.6–1.0	3.7–4.8%
Mask blur	1.0–1.5	5.5–7.5%
0.5 px registration	1.2–1.7	6.5–8.8%
Covariance mismatch	0.7–1.1	4.2–5.6%
Combined mismatch	1.8–2.4	11–15%

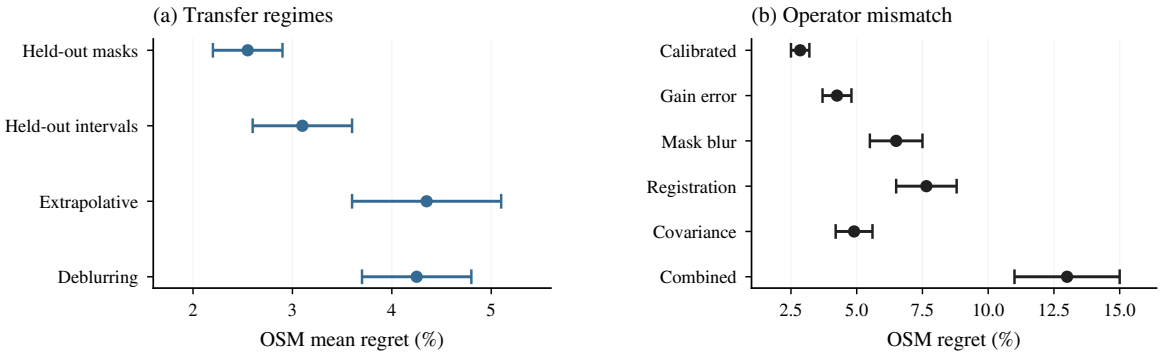


Figure 2. **Transfer and operator-mismatch regret.** Points mark arithmetic midpoints of the reported ranges; whiskers span the range endpoints in Tables 17 and 18. These are not confidence intervals.

from the operator-visible objective.

20. Real SCI Evidence

Public real SCI examples such as Domino and Water Balloon [Qiao et al. 2020] lack aligned ground-truth high-speed video. Quantitative real-measurement reconstruction results are not reported; controlled-simulation transfer therefore does not establish transfer to a physical sensor.

21. Native INT8/FP16 Deployment

Native deployment uses INT8/FP16 on a single NVIDIA A100-SXM4-80GB, batch 1, 256×256, Ubuntu 22.04.5, driver 560.35.05, CUDA 12.6 Update 3, cuDNN 9.8.0, TensorRT 10.13.3, PyTorch 2.13.0, torchvision 0.28.0, and Python 3.12.11. Timing uses 100 warmups and 1,000 synchronized measurements with fixed tactics and excludes disk I/O.

The simulated P1 quality results appear in Table 16; the P4 measurements below use native INT8/FP16 stage execution. These independent precision interventions do not establish a matched accuracy–latency frontier.

Table 19. Measured native P4 end-to-end timing at the primary 0.75×-FP16 budget.

Method	p50 (ms)	p95 (ms)	Speedup
FP16 reference	23.6	25.2	1.00×
Uniform INT8	14.1	15.0	1.67×
Global static mixed	17.3	18.5	1.36×
H-aware gradient	16.2	18.0	1.46×
OSM	16.4	17.9	1.44×
Per-operator static	15.9	17.5	1.48×
Oracle	16.3	17.6	1.45×

Table 20. Observed weak or adverse regimes. These regimes delimit the mechanism and show that the empirical ordering is not uniformly favorable.

Regime	Observed behavior	Consequence
Near-boundary transitions	3–6° localization error; transient 6–8% regret	Transition-localization uncertainty limits hard policy switching; hysteresis is an untested design option.
Stage interactions	Additive $R^2 = 0.84$; pairwise $R^2 = 0.93$	Stage scores are useful but not an exact additive decomposition.
Extrapolative operators	$\tau = 0.64$–0.72; regret 3.6–5.1%	Transfer degrades outside the calibration support.
Combined operator mismatch	Schedule distance 1.8–2.4 stages; regret 11–15%	Precision selection does not compensate for a wrong forward model.
QAT	WORI 0.23; $\tau = 0.77$; regret 2.3%	Lower damage can coincide with flatter/reordered sensitivity.
Fixed-sensor per-operator search	1.7% regret, 15.9 ms p50 vs. OSM 2.7%, 16.4 ms	Cached per-operator calibration is preferable for sufficiently long fixed-H workloads.
Null-space / nonlinear sensing	D_H can miss null-space damage; clipping breaks the linear likelihood model	Pair D_H with reconstruction diagnostics and restrict the clean theory to calibrated linear sensing.

The steady-state timing includes sensing $H/H^\top$, quantization/conversion, policy decision, precision dispatch, and synchronization. Uniform INT8 is faster than OSM; per-operator static is also faster after its search is amortized. P1 reconstruction quality and P4 native timing are distinct regimes and are not combined into a synthetic Pareto plot.

21.1. Measured overhead decomposition

The native measurements distinguish 16.4 ms cached per-snapshot inference from 37.8 ms median new-operator scoring/setup. The two costs occur at different frequencies. Policy-scoring/setup and cached execution are separate measured costs; the percentage decomposition across individual implementation components is not evaluated here.

21.2. Schedule realization and break-even

The 16.4 ms OSM p50 is steady-state execution of an already-realized schedule. The 37.8 ms new-H measurement is policy scoring/setup for a newly calibrated operator; TensorRT engine construction is not included in that measurement.

For a previously unseen calibrated H, OSM score/setup is 37.8 ms median (46.5 ms p90). Exact per-operator calibration evaluates all 256 schedules on a fixed stratified 64-clip calibration subset and costs 286 s median (IQR 249–341 s). Cached inference is 16.4 ms for OSM and 15.9 ms for per-operator static. The measured query-count crossover is 5.6×10^5 (IQR 4.7–6.8×10^5), about 5.2 h at 30 eight-frame SCI snapshots/s, or about 41.5 h at 30 input video frames/s. These are illustrative conversions that exclude engine build/loading and assume a particular exhaustive-search competitor; they are not general device deployment guarantees.

22. OSM Allocation and Schedule Semantics

For equal cost, OSM selects the K largest $s_j(H)$. For unequal measured costs,

$$\hat{z}(H) = \arg\max_{z \in \mathcal{Z}_C} z^\top s(H), \tag{14}$$

where $\mathcal{Z}_C$ contains schedules satisfying the measured budget. Schedule IDs use a deterministic eight-bit stage mapping. Deterministic tie-breaking is kept separate from numerical oracle equivalence.

23. Negative Controls and Failure Modes

The weaker regimes in Table 20 delimit the observed operating conditions.

Table 21. Principal claims and the supplementary evidence associated with each.

Claim	Evidence in this supplement
Operator-conditioned reordering	Five-seed WORI, frozen/full τ, continuous crossings, same-normal invariance.
Static-policy limitation	Exact 256-schedule lattice, $K = 3$ baseline comparison, exact D_H oracle and nonzero static regret.
No-new-search transfer	Held-out masks, held-out intervals, extrapolation, and non-blind deblurring.
Robustness	Calibration-size sweep, multi-reference stability, noise, PTQ/QAT, precision-object comparison, and mismatch conditions.
Systems relevance	Native p50/p95 timing, overhead decomposition, policy scoring/setup, per-operator search cost, and measured break-even.
Scope	Null-space semantics, mismatch limits, saturation/nonlinearity, interaction residuals, and fixed-sensor per-operator wins.

24. Operator Sampling, Boundary Estimation, and Uncertainty

Aggregate WORI and regret depend on the declared operator sampling prior. Policy boundaries are estimated by local interpolation/root finding and measured with fresh samples on both sides. Uncertainty sources are separated conceptually into latent-scene, model-seed, operator-calibration, and finite-operator-sampling components. Hysteresis is a deployment policy around the calibrated operator, not part of the OSM score definition.

25. Robustness Scope

The theory assumes calibrated linear sensing and a declared covariance. Noise robustness is evaluated after whitening. Severe calibration error, nonlinear clipping/saturation, and spatially varying operator drift lie outside the simplest global stage-allocation model. The method should not be interpreted as guaranteeing semantic correctness of the high-precision teacher, because null-space content can remain unconstrained.

26. Distributional Summary

Held-out mask, held-out interval, extrapolative and deblurring experiments are distinct evaluation regimes (Table 17). We observe non-monotonic calibration trends, weaker seeds, boundary misses, and cases where per-operator calibration outperforms OSM.

27. Limitations

First, D_H is teacher-relative and null-space blind. Second, the clean theory assumes calibrated linear sensing and known covariance. Third, susceptibility depends on model, quantizer, intervention context, and calibration distribution. Fourth, schedule interactions prevent exact additivity. Fifth, low-rank basis storage can be nontrivial. Sixth, native gains depend on compiler/kernel/engine realization. Seventh, per-operator calibration can beat OSM for long-lived fixed sensors. Finally, the residual-alignment term prevents interpreting the second-moment score as an exact singleton gain.

28. Claim-to-Evidence Cross-Reference

Table 21 locates the analyses supporting the paper's principal conclusions. The controlled exact-lattice study is the basis for the policy comparison; native INT8/FP16 timings and transfer summaries describe separate evaluation settings.

29. Reproducibility

We report exact-lattice evaluation settings, aggregate results, and hardware specifications. The supplementary tables give per-seed and condition-level results; per-scene/per-schedule reconstruction outputs are not included.

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